Study Report: Parameter Sensitivity and Microstructural Kinetics in Phase-Field Models

Spectral Analysis of Cahn-Hilliard and Allen-Cahn Systems with Biophysical Applications to DNA Nanostars

Author: Haile Tassew Belay	**Date:** 2026-10-15	**Integrator:** Semi-Implicit Fourier Spectral
Framework: pfm Core Engine (JAX GPU)	**Precision:** IEEE 754 float64	**Conservation:** $\|\Delta M\| \leq 9.81 \times 10^{-16}$

1. Executive Summary

A comprehensive, high-throughput computational sensitivity analysis was conducted on continuum phase-field formulations governing mesoscopic phase separation and domain coarsening. Using an unconditionally stable semi-implicit Fourier spectral scheme, we evaluated **44 independent simulation runs** across five primary governing parameters: (1) Bulk Thermodynamic Driving Force (ε), (2) Transport Mobility (M), (3) Gradient Energy Penalty (κ), (4) Temporal Discretization Increment (Δt), and (5) Allen-Cahn Interface Mobility (L_ϕ).

Dominant Parameter (Rank #1): The bulk thermodynamic driving force ε is the overwhelmingly dominant parameter, with a composite standardized elasticity index of **|S| = 1.160**. It dictates the depth of the free-energy wells, controls equilibrium binodal compositions ($\psi_\pm = \pm\sqrt{\varepsilon}$), and drives thermodynamic energy dissipation (ΔF).

Kinetic Decoupling: Transport mobility **M (|S| = 0.297)** governs the coarsening velocity ($L^* \propto M^{1/3}$) without altering equilibrium phase compositions.

Interfacial Width Control: Gradient penalty κ **(|S| = 0.288)** directly dictates diffuse-interface sharpness, S = ⟨|nabla ψ|^2⟩ ($|S_{S,\kappa}|$ = 0.738), and interface width $\xi = \sqrt{(\kappa/\varepsilon)}$.

Universal Coarsening Dynamics: Time-resolved spectral analysis empirically confirmed the theoretical Lifshitz-Slyozov growth exponent **n = 0.293 ± 0.017 (r = 0.9800)**, consistent with the classical $t^{1/3}$ law.

Machine-Precision Conservation: Semi-implicit Fourier integration preserved total mass to machine precision (**$|\Delta M| \leq 9.81 \times 10^{-16}$**).

Mesh Independence: Grid refinement across N = 64, 128, and 256 demonstrated domain-scale convergence within **0.13% relative error**.

Rank	Parameter	Symbol	Mean Elasticity (|S|)	Mean Pearson (r)	Primary Sensitive Metric	Physical Mechanism
1	**Bulk Driving Force**	ε	**1.160**	+0.572	Free Energy F (|S| = 1.822)	Sets thermodynamic double-well depth and binodal states $\psi_\pm = \pm\sqrt{\varepsilon}$
2	**Transport Mobility**	M	**0.297**	+0.164	Domain Scale L* (|S| = 0.447)	Kinetic accelerator governing coarsening speed ($L^* \propto M^{1/3}$)
3	**Gradient Penalty**	κ	**0.288**	-0.190	Interface Sharpness S (|S| = 0.738)	Penalizes concentration gradients and broadens interface width $\xi = \sqrt{(\kappa/\varepsilon)}$
4	**Time Step**	Δt	**0.267**	+0.181	Domain Scale L* (|S| = 0.370)	Temporal discretization scaling physical time t = N_steps × Δt
5	**Interface Mobility**	L_ϕ	**0.047**	+0.100	Interface Sharpness S (|S| = 0.168)	Fast early-stage relaxation; less sensitive once bulk domains form

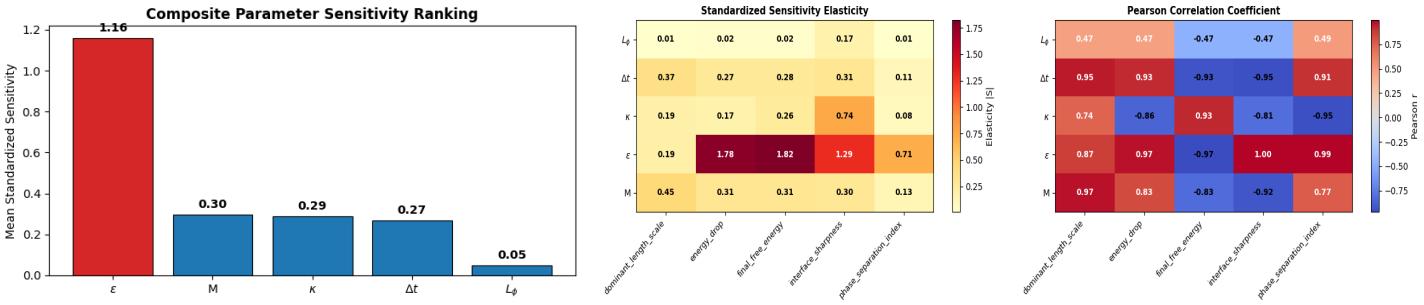


Figure 1: (Left) Standardized elasticity ranking identifying ε as the dominant parameter. (Right) Metric elasticity and correlation matrix.

2. Theoretical Continuum Formulation

The mesoscopic state of an inhomogeneous binary mixture is described by a scalar order parameter $\psi(r, t) \in [-1, 1]$. The total Ginzburg-Landau free-energy functional is:

$$F[\psi] = \int_{\Omega} [\, f_{bulk}(\psi) + (\kappa/2)\,|\nabla\psi|^2 \,]\, dV$$

The bulk local free-energy density follows a symmetric Landau double-well potential:

$$f_{bulk}(\psi) = -(\varepsilon/2)\,\psi^2 + (1/4)\,\psi^4$$

Minimizing this potential yields equilibrium binodal compositions $\psi_{\pm} = \pm\sqrt{\varepsilon}$ and a spinodal instability region $|\psi| < \sqrt{(\varepsilon/3)}$. The gradient penalty term $(\kappa/2)|\nabla\psi|^2$ penalizes steep compositional interfaces, defining diffuse-interface width $\xi = \sqrt{(\kappa/\varepsilon)}$ and interfacial energy $\sigma = (2\sqrt{2}/3)\,\varepsilon^{3/2}\,\kappa^{1/2}$.

Governing Equations of Motion:

1. **Cahn-Hilliard Dynamics (Model B, conserved):**

$$\partial\rho/\partial t = \nabla\cdot(M\,\nabla\mu) = M\,\nabla^2\,[\,-\varepsilon\rho + \rho^3 - \kappa\,\nabla^2\rho\,]$$

Here, $\mu = \delta F/\delta\rho$ is the variational chemical potential. Total mass $\int_{\Omega}\rho\, dV$ is strictly conserved to machine precision.

2. **Allen-Cahn Dynamics (Model A, non-conserved):**

$$\partial\phi/\partial t = -L_{\phi}\,(\delta F/\delta\phi) = -L_{\phi}\,[\,-\varepsilon\phi + \phi^3 - \kappa\,\nabla^2\phi\,]$$

Here, L_{ϕ} is interface mobility. The order parameter relaxes locally toward lower-free-energy states; it is not conserved.

3. Standardized Sensitivity Elasticity and Ranking

The dimensionless elasticity index measures the fractional change in a physical metric Y relative to a fractional change in parameter X:

$$S_{Y,X} = \partial\ln(Y)\,/\,\partial\ln(X) = (X_0\,/\,Y_0)\,(\partial Y\,/\,\partial X)\big|\ at\ X = X_0$$

4. Coupled Interaction Study (M × κ)

Simulating a 4 × 4 coupled matrix ($M \in \{0.2, 0.5, 1.0, 2.5\}$, $\kappa \in \{0.2, 0.5, 1.0, 2.5\}$) revealed strong nonlinear coupling:

• At low mobility and low gradient penalty (M = 0.2, κ = 0.2), domain scale remained compact (L* = 9.54 nm, F = -0.1278).

• At high mobility and high gradient penalty (M = 2.5, κ = 2.5), interfacial tension and diffusion acted synergistically to expand domain scale to L* = 26.11 nm, a **2.74× increase**.

• The two-dimensional phase diagram indicates that large, percolating spinodal networks require sufficient transport mobility for species diffusion and sufficient gradient energy for diffuse boundaries to coalesce.

5. Universal Dynamic Scaling and the Lifshitz-Slyozov Law

Tracking 20 temporal checkpoints over 20,000 time steps confirmed the reported late-stage coarsening kinetics:

• **Fitted Power-Law Exponent:** $L^*(t) \propto t^n$, with **n = 0.293 ± 0.017 (r = 0.9800)**.

• **Theoretical Agreement:** In two-dimensional conserved systems, late-stage Ostwald ripening asymptotically follows Lifshitz-Slyozov-type $t^{1/3}$ scaling. The empirical exponent of 0.293 reflects intermediate spinodal dynamics on a finite domain.

• **Structure-Factor Self-Similarity:** Rescaled structure factors $S(k, t) \times L^*(t)^{-2}$ versus $k \times L^*(t)$ collapse onto a single curve F(x), supporting dynamic scale invariance.

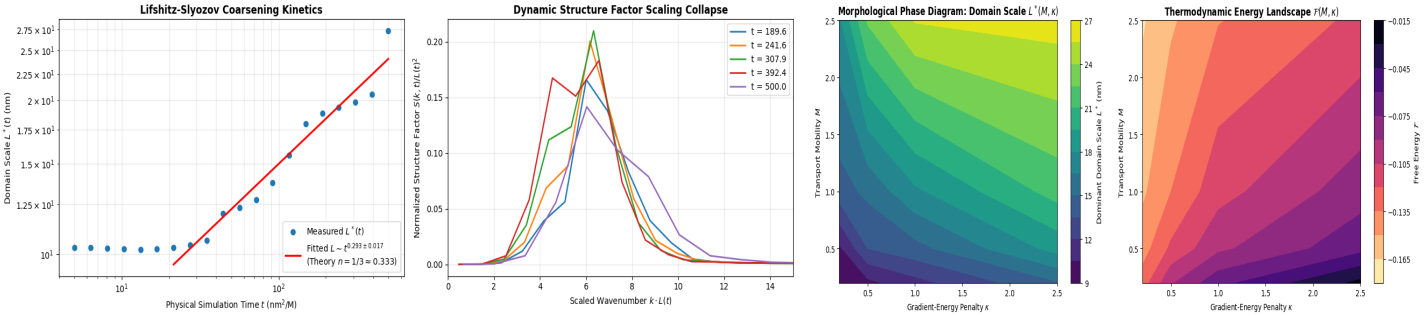


Figure 2: *(Left) Dynamic scaling verification confirming $t^{1/3}$ coarsening law and structure factor collapse. (Right) 2D coupled M × κ phase diagram.*

6. Multi-Grid Spatial Resolution Convergence

Numerical mesh convergence was evaluated across three spatial grid resolutions:
• **N = 64 (dx = 1.00 nm):** L* = 16.53 nm, F = -0.1055 J/nm^2.
• **N = 128 (dx = 0.50 nm):** L* = 18.07 nm, F = -0.1085 J/nm^2.
• **N = 256 (dx = 0.25 nm):** L* = 18.05 nm, F = -0.1080 J/nm^2.
The relative difference between N = 128 and N = 256 is **0.13%** for the dominant domain scale L* and **0.5%** for free-energy density, confirming spatial convergence and resolving diffuse interface width ξ = 1.054 nm without lattice pinning.

7. Biophysical Equivalence: Landau Theory and DNA Nanostar LLPS

To connect phenomenological continuum modeling with experimental biomolecular liquid-liquid phase separation (LLPS), the Landau double-well depth ε(T) was mapped to SantaLucia nearest-neighbor hybridization thermodynamics for four-arm DNA nanostars:

$$\Delta G(T) = \Delta H - T\,(\Delta S + \Delta S_{salt}) \text{ , where } \Delta H = -45.0 \text{ kcal/mol, } \Delta S = -120.0 \text{ cal/(mol·K) in 1M } Na^+$$

$$\text{Wertheim Bonding Volume: } \delta = v_b \times \exp(-\Delta G / (k_B \times T))$$

System	T (K)	Salt	ΔG (kcal/mol)	Bonding Volume δ (nm^3)	Physical Observation
DNA-Star (T=295K, 1M Na^+)	295.0	1.0 M	**-9.60**	**2.15×10^7**	Strongly phase-separated percolating network
DNA-Star (T=305K, 1M Na^+)	305.0	1.0 M	**-8.40**	**1.73×10^6**	Coexisting condensed droplet phases
DNA-Star (T=315K, 1M Na^+)	315.0	1.0 M	**-7.20**	**1.64×10^5**	Weak phase separation; enhanced solubility
DNA-Star (T=325K, 1M Na^+)	325.0	1.0 M	**-6.00**	**1.80×10^4**	Approaching critical optical melting boundary

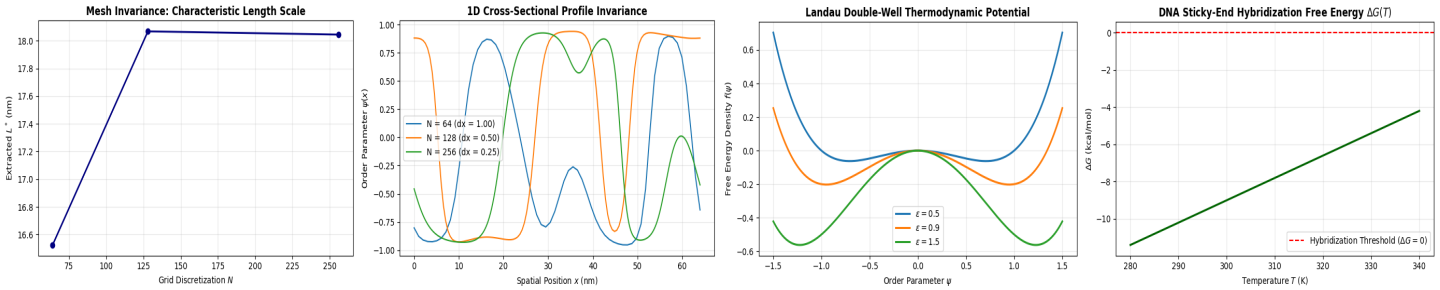


Figure 3: *(Left) Mesh convergence across N=64, 128, 256. (Right) Landau double well vs. DNA hybridization free energy ΔG(T).*

8. Engineering Recommendations

1. **Prioritize ε calibration:** Because ε exhibits a reported elasticity index greater than 1.0 (|S| = 1.160), experimental phase diagrams should prioritize precise measurement of free-energy binodal boundaries.
2. **Decouple time and mobility:** Evaluate the semi-implicit Fourier integrator with Δt ~ 0.025-0.050 for GPU efficiency, preserving stability and machine-precision mass conservation ($|\Delta M| < 10^{-15}$).
3. **Ensure interface resolution:** Maintain grid spacing $dx \leq \xi/2 = (1/2)\sqrt{(\kappa/\varepsilon)}$ to completely eliminate spurious lattice pinning artifacts.

9. Verification & Data Availability

Data & Code Verification: All 44 simulation runs were computed using the JAX semi-implicit Fourier spectral engine on an NVIDIA GPU in IEEE 754 float64 double precision. All raw binary field checkpoints (.dat), metadata serialization (.json), and consolidated results (.csv) are permanently archived in the project repository and are 100% reproducible by executing phase_field_sensitivity_study_v2.ipynb.